

1. Consider the quantity

$$\begin{aligned}\frac{f(z + \Delta z) - f(z)}{\Delta z} &= \frac{(z + \Delta z)^3 - z^3}{\Delta z} \\ &= \frac{z^3 + 3z^2\Delta z + 3z(\Delta z)^2 + (\Delta z)^3 - z^3}{\Delta z} \\ &= 3z^2 + 3z\Delta z + (\Delta z)^2\end{aligned}$$

Next consider its limit as $\Delta z \rightarrow 0$. The limit of a sum of a finite number of terms exists, and is equal to the sum of limits, provided the limits of terms exist individually. This is the case above and therefore

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = 3z^2$$

2. (a) $f(z) = \operatorname{Re} z = x$. At a given z , $f(z)$ varies with x , i.e. along the direction $y = \text{const}$, but is constant along $x = \text{const}$. Thus it is not differentiable (in the sense of differentiability in complex analysis) at any z . Hence, it is nowhere analytic.
- (b) If we write z in polar form, $z = r \exp(i\theta)$, then $f(z) = |z|^2 = r^2$. At a given z , $f(z)$ varies along the radial direction, i.e. along the ray $\theta = \text{const}$, but is constant along the circle $r = \text{const}$. Thus, it is nowhere analytic. (Above we have not worried about the point $z = 0$, but a function cannot be analytic at an isolated point, so the conclusion is not affected.)
- (c) Writing $z = r \exp(i\theta)$, for $r > 0$:

$$f(z) = \frac{z}{z^*} = \exp(2i\theta)$$

$f(z)$ is constant along the ray $\theta = \text{const}$, but varies along the circle $r = \text{const}$. It is, further, undefined at $z = 0$. It is nowhere analytic.

3. If $f(z) = u(x, y) + i v(x, y)$, then the Cauchy-Riemann equations are

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

where subscripts denote partial derivatives.

$$\begin{aligned}\text{(a) } f(z) = z^3 &= (x + iy)^3 = (x^3 - 3xy^2) + i(3x^2y - y^3) \\ \Rightarrow u(x, y) &= x^3 - 3xy^2 \quad \text{and} \quad v(x, y) = 3x^2y - y^3 \\ u_x &= 3x^2 - 3y^2 \quad v_y = 3x^2 - 3y^2 \\ u_y &= -6xy \quad v_x = 6xy\end{aligned}$$

The Cauchy-Riemann equations are satisfied for all z . $f(z) = z^3$ is everywhere analytic.

$$(b) \quad f(z) = \frac{1}{z} = \frac{1}{x + iy} = \frac{x - iy}{x^2 + y^2}$$

$$\Rightarrow \quad u(x, y) = \frac{x}{x^2 + y^2} \quad \text{and} \quad v(x, y) = -\frac{y}{x^2 + y^2}$$

$$\begin{aligned} u_x &= \frac{1}{x^2 + y^2} - \frac{2x^2}{(x^2 + y^2)^2} = \frac{-x^2 + y^2}{(x^2 + y^2)^2} \\ v_y &= -\frac{1}{x^2 + y^2} + \frac{2y^2}{(x^2 + y^2)^2} = \frac{-x^2 + y^2}{(x^2 + y^2)^2} \\ u_y &= -\frac{2xy}{(x^2 + y^2)^2} \quad v_x = \frac{2xy}{(x^2 + y^2)^2} \end{aligned}$$

The Cauchy-Riemann equations are satisfied for all z , except $z = 0$ which is a singular point of $1/z$. $f(z) = 1/z$ is everywhere analytic except at the origin.

$$(c) \quad f(z) = \operatorname{Re} z = x$$

$$\Rightarrow \quad u(x, y) = x \quad \text{and} \quad v(x, y) = 0$$

$$u_x = 1 \quad v_y = 0 \quad u_y = 0 \quad v_x = 0$$

The Cauchy-Riemann equations are not satisfied anywhere. $f(z) = \operatorname{Re} z$ is nowhere analytic.

$$(d) \quad f(z) = \exp z = \exp x \cos y + i \exp x \sin y$$

$$\Rightarrow \quad u(x, y) = \exp x \cos y \quad \text{and} \quad v(x, y) = \exp x \sin y$$

$$\begin{aligned} u_x &= \exp x \cos y & v_y &= \exp x \cos y \\ u_y &= -\exp x \sin y & v_x &= \exp x \sin y \end{aligned}$$

The Cauchy-Riemann equations are satisfied for all z . $f(z) = \exp z$ is everywhere analytic.

$$(e) \quad f = z^* = x - iy$$

$$\Rightarrow \quad u(x, y) = x \quad \text{and} \quad v(x, y) = -y$$

$$u_x = 1 \quad v_y = -1 \quad u_y = 0 \quad v_x = 0$$

The Cauchy-Riemann equations are not satisfied anywhere. $f(z) = z^*$ is nowhere analytic.

$$(f) \quad f(z) = \sin z = \sin x \cosh y + i \cos x \sinh y$$

$$\Rightarrow \quad u(x, y) = \sin x \cosh y \quad \text{and} \quad v(x, y) = \cos x \sinh y$$

$$\begin{aligned} u_x &= \cos x \cosh y & v_y &= \cos x \cosh y \\ u_y &= \sin x \sinh y & v_x &= -\sin x \sinh y \end{aligned}$$

The Cauchy-Riemann equations are satisfied for all z . $f(z) = \sin z$ is everywhere analytic.

$$(g) \quad f(z) = \cosh z = \cosh x \cos y + i \sinh x \sin y$$

$$\Rightarrow \quad u(x, y) = \cosh x \cos y \quad \text{and} \quad v(x, y) = \sinh x \sin y$$

$$\begin{aligned} u_x &= \sinh x \cos y & v_y &= \sinh x \cos y \\ u_y &= -\cosh x \sin y & v_x &= \cosh x \sin y \end{aligned}$$

The Cauchy-Riemann equations are satisfied for all z . $f(z) = \cosh z$ is everywhere analytic.

4. We can write

$$\begin{aligned} \cos z &= \frac{\exp(iz) + \exp(-iz)}{2} \\ \sin z &= \frac{\exp(iz) - \exp(-iz)}{2i} \end{aligned}$$

as

$$\begin{aligned} 2 \cos z &= \exp(iz) + \exp(-iz) \\ 2i \sin z &= \exp(iz) - \exp(-iz) \end{aligned}$$

Adding and subtracting the last two equations:

$$\exp(\pm iz) = \cos z \pm i \sin z$$

In an analogous way, we can write

$$\begin{aligned} \cosh z &= \frac{\exp z + \exp(-z)}{2} \\ \sinh z &= \frac{\exp z - \exp(-z)}{2} \end{aligned}$$

as

$$\exp(\pm z) = \cosh z \pm \sinh z$$

(a)

$$\begin{aligned} \sin(z_1 + z_2) &= \frac{1}{2i} [\exp(i(z_1 + z_2)) - \exp(-i(z_1 + z_2))] \\ &= \frac{1}{2i} [\exp(iz_1) \exp(iz_2) - \exp(-iz_1) \exp(-iz_2)] \\ &= \frac{1}{2i} [(\cos z_1 + i \sin z_1)(\cos z_2 + i \sin z_2) \\ &\quad - (\cos z_1 - i \sin z_1)(\cos z_2 - i \sin z_2)] \\ &= \sin z_1 \cos z_2 + \cos z_1 \sin z_2 \end{aligned}$$

(b)

$$\begin{aligned}\cosh(z_1 + z_2) &= \frac{1}{2} [\exp(z_1 + z_2) + \exp(-(z_1 + z_2))] \\ &= \frac{1}{2} [\exp(z_1) \exp(z_2) + \exp(-z_1) \exp(-z_2)] \\ &= \frac{1}{2} [(\cosh z_1 + \sinh z_1)(\cosh z_2 + \sinh z_2) \\ &\quad + (\cosh z_1 - \sinh z_1)(\cosh z_2 - \sinh z_2)] \\ &= \cosh z_1 \cosh z_2 + \sinh z_1 \sinh z_2\end{aligned}$$

(c)

$$\begin{aligned}\sin(iz) &= \frac{\exp(i(iz)) - \exp(-i(iz))}{2i} \\ &= \frac{1}{i} \frac{\exp(-z) - \exp z}{2} \\ &= -i \frac{\exp(-z) - \exp z}{2} = i \sinh z\end{aligned}$$

(d)

$$\cosh(iz) = \frac{\exp(iz) + \exp(-iz)}{2} = \cos z$$

$$\begin{aligned}\sin(i\pi/2) &= i \sinh(\pi/2) \\ \sinh(i\pi/2) &= \frac{1}{i} \sin(ii\pi/2) = i \sin(\pi/2) = i \\ \cos(i\pi) &= \cosh(ii\pi) = \cosh(-\pi) = \cosh \pi \\ \cosh(i\pi) &= \cos \pi = -1\end{aligned}$$

5. (a)

$$\begin{aligned}\operatorname{Log}(1 + i) &= \log |1 + i| + i \operatorname{Arg}(1 + i) \\ &= \log \sqrt{2} + i \frac{\pi}{4}\end{aligned}$$

(b)

$$\begin{aligned}\operatorname{Log}(2 - 2i) &= \log |2 - 2i| + i \operatorname{Arg}(2 - 2i) \\ &= \log(2\sqrt{2}) - i \frac{\pi}{4}\end{aligned}$$

(c)

$$\begin{aligned}\operatorname{Log}(-5) &= \log 5 + i \operatorname{Arg}(-5) \\ &= \log 5 + i\pi\end{aligned}$$

6. (a) $z = \log(-e^2)$. Using $\log z = \log |z| + i \arg z$:

$$\begin{aligned}\log(-e^2) &= \log(e^2) + i(\pi + 2k\pi) \\ &= 2 \log e + i(1 + 2k)\pi \\ &= 2 + i(1 + 2k)\pi\end{aligned}$$

where $k = 0, \pm 1, \pm 2, \dots$

(b) If $\exp z = -1 + i$ then

$$\begin{aligned}z &= \log(-1 + i) = \log |-1 + i| + i \arg(-1 + i) \\ &= \log \sqrt{2} + i(3\pi/4 + 2k\pi) \\ &= \frac{1}{2} \log 2 + i(3\pi/4 + 2k\pi) \quad , \quad k = 0, \pm 1, \pm 2, \dots\end{aligned}$$

(c) If $\exp(iz) = \sin(2i) = i \sinh 2$ then

$$\begin{aligned}iz &= \log(i \sinh 2) = \log \sinh 2 + i(\pi/2 + 2k\pi) \\ \Rightarrow z &= -i \log \sinh 2 + (\pi/2 + 2k\pi) \quad , \quad k = 0, \pm 1, \pm 2, \dots\end{aligned}$$
